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Motor Imagery EEG Discrimination Using the Correlation of Wavelet Features

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Abstract

A novel method for motor imagery (MI) electroencephalogram (EEG) data classification is proposed in this study. Time-frequency representation is constructed by means of continuous wavelet transform from EEG signals and then weighted with 2-sample *t*-statistics, which are also used to automatically select the area of interest in advance. Finally, normalized cross-correlation is used to discriminate the test MI data. Compared with the nonweighted version on MI data, the experimental results indicate that the proposed system achieves satisfactory results in the applications of brain-computer interface (BCI).

Keywords

motor imagery (MI), electroencephalography (EEG), time-frequency representation, two-sample t-statistics, normalized cross correlation, brain-computer interface (BCI)

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Introduction

The goal of BCI is to allow direct transmission of messages by analyzing the brain's mental activities.[1-8] BCI systems based on single-trial analysis of EEG signals associated with MI have become popular in the past decade.[8] Special characteristics of event-related desynchronization (ERD) and synchronization (ERS) in mu and beta rhythms, over the sensorimotor cortices, during MI tasks are used to discriminate between left and right MIs.[9,10] In BCI applications, the performance of mental task classification depends greatly on feature extraction, such as the representation of MI data. Accordingly, it may be a better choice to expand 1-dimensional (1D) signal to 2D time-frequency representation, because of the complexity of MI data to extract important features.

Many studies discriminate MI EEG data by means of ERD/ERS components.[9,10] However, most of them work on 1D temporal data and perform too complicated an operation, which diminishes the performance and practicality. The principal aim of this study is to propose a simple and practical EEG-based BCI system, which extracts important time-frequency features, using 2-sample *t*-statistics from selected continuous wavelet transform (CWT) plots, and discriminates with normalized cross-correlation for MI recognition. The feature extraction/representation, with CWT and 2-sample *t*-statistics, further enhances the performance in BCI applications. The proposed noninvasive MI BCI system can effectively increase the accuracy of imagined motor control for subjects who are disabled.

Feature extraction/representation is an important topic that substantially affects the classification accuracy of MI tasks in BCI work. An effective feature extraction method can obtain good classification results even if the adopted classifier is unsophisticated. Discrete wavelet transform (DWT) is powerful in selecting features from the time-frequency domain[11], and is efficient in ERD/ERS representation. However, it has the drawback of a predefined representation that cannot be adjusted to a given ERD/ERS structure.[12,13] CWT provides a highly redundant representation of EEG signals in the time-scale domain.[12] Hence, it can be applied for the precise localization of ERD/ERS components in the time-scale domain.[13]

Two-sample *t*-statistic is a commonly used technique for assessing whether the means of 2 groups are statistically different from each other. It can be used to detect the most discriminative ERD/ERS components for different mental tasks. In this study, the CWT is used together with 2-sample *t*-statistics. These are also used to automatically select the area of interest, in advance, for feature extraction and representation in the time-frequency domain.

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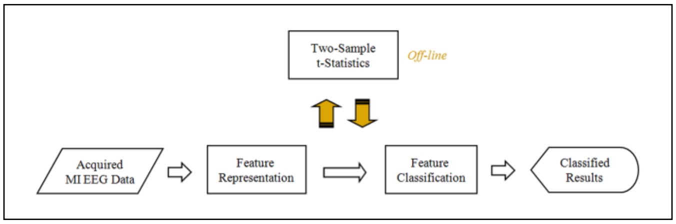

Figure 1. Flowchart of the proposed brain-computer interface system.

To evaluate the performance of the proposed method for MI classification, the pure time-frequency representation/feature, without being weighted with 2-sample t-statistics, is implemented for the comparison. It is worthwhile to note that the 2D time-frequency representation improves upon an 1D temporal feature with regard to the attributes of interest for this study.

Materials and Methods

The proposed EEG discrimination system for single-trial MI classification is illustrated in Figure 1. The procedure consists of data description, feature representation, estimation of 2-sample t-statistics, and feature classification. CWT is used to build 2D time-frequency representation in feature representation. Two-sample t-statistics, used to weight the representation, are also used to choose the appropriate weighting area ahead. Finally, a simple normalized cross-correlation is used to classify test data into various MI states, left or right.

Data Description

The data set was recorded from 5 untrained subjects (4 males and 1 female) in a shielded room using 10 silver/silver chloride electrodes. As illustrated in Figure 2, they consist of 10 scalp EEG channels (C3, C5, FC3, C1, CP3, C4, C2, FC4, C6, and CP4).[14] All electrodes were referenced to the A1 lead at the left earlobe. Before being sampled at the rate of 128 Hz, the EEG data were filtered by an analog band-pass filter with cutoff frequencies at 0.5 and 30 Hz, and amplified by a multiple of 10 000. The task was to control a bar by means of imagined left or right hand movements similar to the Graz BCI.[15,16] Each subject performed 300 trials. No trials were removed during the EEG data processing. No measurement was done with artifact removal. The length of each trial was 8 to 9 seconds. The first 2 seconds of the trial were quiet, then an acoustic stimulus indicated the beginning of the experiment at $t = 2$ seconds, and a fixation cross + was displayed for 1 second; then, at $t = 3$ seconds, an arrow (left or right) was displayed as a cue. (The data recorded between 3 and 8 seconds are considered to be event related.) At the same time, each subject was asked to move a bar by imagining left or right hand movements according to the direction of the cue.

Preprocessing

Non-EEG noise is significantly different from EEG signals in both topographical and frequency characteristics. The mu and beta rhythms of the EEG are those components with frequencies distributed between 8 and 30 Hz and located over the sensorimotor cortex. Electro-oculography signals are maximal at low frequencies (<5 Hz) and are prominently situated over the anterior head regions. Therefore, an appropriate filtering method can increase the signal-to-noise ratio by enhancing EEG signals and reducing non-EEG noise at the same time. The surface Laplacian filter is a simple but effective filtering method.[17] It calculates the second derivative of the spatial voltage distribution for a selected electrode. It is a high-pass spatial filter that enhances localized activities and reduces background noise. This filter is achieved by subtracting the average potential of a set of surrounding electrodes from the electrode of interest. The distance between the selected electrode and its surrounding electrodes demonstrates the characteristic of surface Laplacian filtering that the greater the distance, the greater the insensitivity to highly localized potentials.

Feature Representation

Continuous wavelet transform gives a highly redundant representation of EEG signals in the time-scale domain and can be applied for precise localization of ERD/ERS components through time-frequency analysis.[12,13] The CWTs of EEG data, performing left and right MI in both *C3* and *C4* channels, are represented, respectively, as

$$W_s^{C,n}(j,k) = \int_R f_s^{C,n}(x)\frac{1}{\sqrt{j}}\psi\left(\frac{x-k}{j}\right)dx \quad (1)$$

Where

$$\frac{1}{\sqrt{j}}\psi\left(\frac{x-k}{j}\right)$$

are the dilated and translated versions of the wavelet function $\psi(x)$ at scale j and shift k, and $W_s^{C,n}(j,k)$ stands for the CWT of EEG data $f_s^{C,n}(x)$ in which state s represents either left or right MI, C indicates either channel *C3* or *C4*, and n represents the trial number. $W_s^{C,n}(j,k)$ can be considered a 2D time-frequency representation in which the precise localization of ERD/ERS components[13] is kept. $W_s^{C,n}(j,k)$ is performed with Daubechies wavelets of order 4 in this study[18] since it has the characteristic of being compactly supported with extreme phase and the highest number of vanishing moments for a given support width.

However, the 2D time-frequency representation $W_s^{C,n}(j,k)$ is noisy. It is difficult to locate precisely the discriminant region based on the extreme spectral difference.[19] In

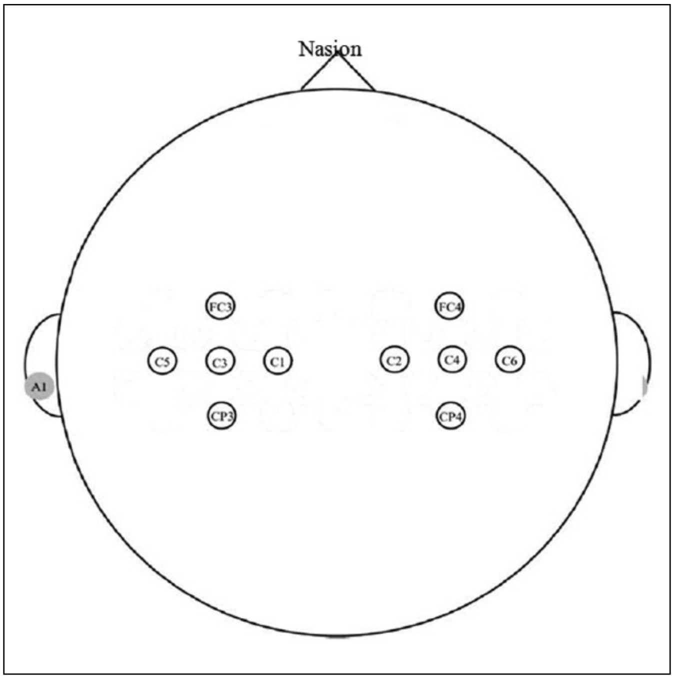

Figure 2. Ten EEG electrode locations in the international standard 10-20 system.

other words, the best discriminant region does not exist at a single point but lies in a particular range of time and scale. Taking the time and scale domains into account, we slightly smooth the power spectrum to reduce the noise. That is, we convolute the representation $W_s^{C,n}(j,k)$ with Gaussian smoothing $G(x,y)$,

$$WG_s^{C,n}(j,k) = W_s^{C,n}(j,k) * G(x,y) \tag{2}$$

In general, using a wider frequency range through acquired EEG signals can achieve higher classification accuracy when compared with a narrower one.[20] In this study, a wide frequency range, including mu and beta rhythmic components, is used to acquire important power spectrums for motor imagery classification. The 2-sample *t*-statistic for the 2 different MIs is subsequently represented as

$$t^C(j,k) = \frac{\left|M_L^C(j,k) - M_R^C(j,k)\right|}{\sqrt{\left(\frac{(N_L-1)\cdot V_L^C(j,k) + (N_R-1)\cdot V_R^C(j,k)}{N_L+N_R-2}\right)\left(\frac{1}{N_L}+\frac{1}{N_R}\right)}} \tag{3}$$

where $M_s^C(j,k)$ and $V_s^C(j,k)$ represent mean and the variance of CWTs for left and right motor imagery in channels *C3* and *C4*, respectively. The denominator in Equation (3) indicates the pooled variance of the 2 states for channel *C*. The expression $t^C(j,k)$ with a different scale *j* and different time *k* for the 2 distinct states, *L* and *R*, can form a 2D representation with respect to both time and frequency. It owns different characteristics in comparison with the original 2D time-frequency representation, $W_s^{C,n}(j,k)$. Points having local maxima in 2D representation $t^C(j,k)$ imply that there is local maximal difference between the 2 states in the time-scale domain. In other words, these 2 EEG signal states are best discriminated at a particular time and scale range.[19] Only the area within a predefined range (2 seconds and 20 Hz), its center being the peak of 2D representation $t^C(j,k)$, is used to reduce computation costs in this study.

Feature Representation With Weighting

$M_s^C(j,k)$ and $t^C(j,k)$ are extracted for each channel and each state. Instead of considering 2 representations for channels *C3* and *C4* individually, we combine $M_s^{C_3}(j,k)$ and $M_s^{C_4}(j,k)$ together to form the new $M_s(j,k)$, and $t(j,k)$ is generated by combining the representations $t^C(j,k)$ of both channels *C3* and *C4* together.

During the training, left and right characteristic time-frequency representations (CTFRs) are obtained by normal averaging, respectively, and they are then combined together. However, CTFRs at different time, frequency and channel positions may not contribute equally to improving classification accuracy. Some positions may even lead to incorrect classification because of the complexity and variational characteristics in ERD/ERS. Accordingly, it is necessary to refine CTFRs by excluding the misleading time-frequency-channel regions that may cause the reduction in classification accuracy. The CTFRs are refined by weighting, with the representation $t(j,k)$,

$$tCTFR(j,k) = CTFR(j,k) * t(j,k) \tag{4}$$

Finally, we obtain the *t*-statistic-weighted characteristic time-frequency representations (tCTFRs) by multiplying the CTFRs with their joint weight $t(j,k)$. The tCTFRs are subsequently used to evaluate the classification accuracy of the test data. To reduce computation costs and increase classification accuracy, only the data within the area of interest are used for MI classification.

Classification

During the testing, the test trial is transformed into time-frequency representation by Equations (1) and (2), and it is then weighted by the representation $t(j,k)$ to obtain the processed test trial *tT*. Given the left and right tCTFRs, a *tT* is classified as either left or right MI by evaluating the normalized cross-correlation (NCC) between the given processed test trial *tT* and the respective tCTFRs tC_s, tC_L or tC_R,

$$NCC(tT, tC_s) = \frac{Cov(tT, tC_s)}{Std(tT)Std(tC_s)} \tag{5}$$

where $Cov(\cdot)$ and $Std(\cdot)$ denote the covariance and standard deviation, respectively. In statistics, NCC is a measure of linear dependence between two variables. It produces the value between −1 and 1 in which 1 is total positive correlation and −1 is total negative correlation. NCC is used to measure the degree of linear dependence between 2 variables. In addition, NCC is invariant (up to a sign) to separate changes in location and scale between 2 variables. Accordingly, correlation will succeed even if the 2D time-frequency representation varies with position. The range is independent of the size of 2D time-frequency representation. NCC is invariant to changes in the amplitude of time-frequency representation. NCC, which is better than other sophisticated techniques, is used to overcome these limitations. Additionally, it can be performed fast and efficiently for real-time applications. Finally, the classification result of the processed test trial tT is obtained,

$$f(tT) = \mathrm{sgn}\left(NCC\left(tT, tC_L\right) - NCC\left(tT, tC_R\right)\right) \quad (6)$$

If $f(\cdot)$ is a positive number, the trial is classified as left MI. Otherwise, it is classified as right.

Experimental Results

Approaches of Performance Evaluation

The classification tests for MI data are carried out using tenfold cross validation in all experiments. In addition to classification accuracy, the receiver operating characteristics (ROC) curve is another popular approach to evaluate the performance of a method. The ROC is a plot of the true positive rate against the false positive rate for every possible cutoff of a diagnostic test. The performance can be measured by the area under the ROC curve (AUC), which is used to describe the probability of random samples being assigned to the correct class. In this study, we evaluate the performance of all the experiments by means of both the classification accuracy and AUC.

Classification Accuracy of Feature Representations

To assess the performance of the proposed method, a well-known time-frequency approach is also implemented for comparison[21] in addition to the nonweighted feature representation. It is proposed to use wavelet-packet features selected with a genetic algorithm. Table 1 lists the comparisons of classification accuracy among wavelet-packet analysis, nonweighted and weighted feature representations. The second is weighted by means of 2-sample t-statistics, while the third is not. These 3 representations are compared at the same data for each subject. In other words, the listed values demonstrate only the deviations of performance among different feature representations. The average classification accuracies for the wavelet-packet analysis, nonweighted and weighted feature representations, are 75.6%, 78.3%, and 83.5%, respectively. We observe that the weighted feature representation gives the best results for all subjects. The results indicate that the weighted feature representation obtains the promising results in MI classification.

Area Under the Receiver Operating Characteristic Curve of Feature Representations

An experiment was also performed to estimate AUCs of different feature representations. The comparison of AUC among wavelet-packet analysis, nonweighted and weighted feature representations is listed in Table 2. Data used to compare among these three representations are also the same, and the resulting values only exhibit their deviations of AUC. The average accuracy for wavelet-packet analysis is 0.72, whereas the accuracies for nonweighted and weighted feature representations are 0.76 and 0.82, respectively. The results denote that the weighted feature representation gives the best results for all subjects. The results also show that the weighted feature representation is better in terms of the AUC measure.

Discussion

The purpose of this study is to propose a reliable feature representation with weighting to enhance the MI EEG classification. The measure of improvement is via increased recognition rate and higher AUC. Time-frequency–based features have been investigated[11-13] for feature representation, and shown to be effective in improving classification accuracy. These results suggest that the analysis of ERD/ERS components in the time-frequency domain may contribute to more accuracy in MI classification.

Statistical Evaluation of Feature Representations in Classification Accuracy

The comparison of classification accuracy among wavelet-packet analysis, nonweighted and weighted feature representations, is listed in Table 1. The nonweighted and weighted feature representations are both 2D time-frequency features, but their contents are different. The weighted feature representation is weighted via 2-sample t-statistics to refine the significant information and simultaneously remove the noise, while the former is not. Experimental results reveal that the latter is better in terms of classification accuracy for all subjects. Two-way analysis of variance and multiple comparison tests are performed to validate whether these 3 feature representations are significantly different or not. The results indicate that the difference of classification accuracy “between wavelet-packet analysis and weighted feature representation” and “between the nonweighted and weighted feature representations” are both significant (P = .0019 and P = .0068, respectively). Accordingly, feature representation with t-statistics-weighting can further find helpful features and remove the noise at the same time, which can effectively increase the accuracy of imagined motor control.

Table 1. Comparison of Classification Accuracy Among Wavelet-Packet Analysis, Nonweighted and Weighted Feature Representations.

Classification Accuracy	Wavelet-Packet Analysis (%)	Feature Representation (%)	Weighted Feature Representation (%)
S1	82.1	81.7	89.3
S2	81.0	83.3	87.7
S3	76.4	78.9	83.0
S4	70.8	75.1	77.5
S5	67.6	72.4	79.8
Average	75.6	78.3	83.5

Table 2. Comparison of Area Under the Receiver Operatic Characteristic Curve (AUC) Among Wavelet-Packet Analysis, Nonweighted and Weighted Feature Representations.

AUC	Wavelet-Packet Analysis	Feature Representation	Weighted Feature Representation
S1	0.74	0.77	0.86
S2	0.81	0.80	0.86
S3	0.68	0.73	0.84
S4	0.70	0.73	0.75
S5	0.68	0.75	0.81
Average	0.72	0.76	0.82

Statistical Evaluation of Feature Representations in Area Under the Receiver Operating Characteristic Curve

In addition to classification accuracy, the ROC curve is another approach in evaluating the performance. The performance is measured by the AUC, which is used to describe the probability of random samples being assigned to the correct class. Table 2 lists the comparison of AUC among wavelet-packet analysis, nonweighted and weighted feature representations. Two-way analysis of variance and multiple comparison tests are performed again to verify if these 3 feature representations are significantly different. The results denote that there are significant differences "between wavelet-packet analysis and weighted feature representation" (P = .0101) and "between the nonweighted and weighted feature representations" (P = .0113). Therefore, they also reveal that feature representation with t-statistics-weighting is a better choice than the other 2 in BCI applications.

Normalized cross-correlation is a simple approach and is used to analyze the MI EEG data in the experiments. Its simplicity enables the performance of feature representations to be clearly demonstrated, which is the primary focus of this study.

Conclusion

In this study, we propose a novel EEG discrimination system for single-trial MI classification. Continuous wavelet transform is used to construct time-frequency representation, which provides time-frequency localization of ERD/ERS components. The representation is then weighted with 2-sample t-statistics, which extract the information that is most discriminative after the area of interest is selected. Finally, NCC is used for the discrimination of test MI data. The experimental results demonstrate that the proposed method is promising in the classification of mental tasks. It offers a potential benefit when using 2D time-frequency features in BCI projects.

Declaration of Conflicting Interests

The author(s) declared no conflicts of interest with respect to the research, authorship, and/or publication of this article.

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